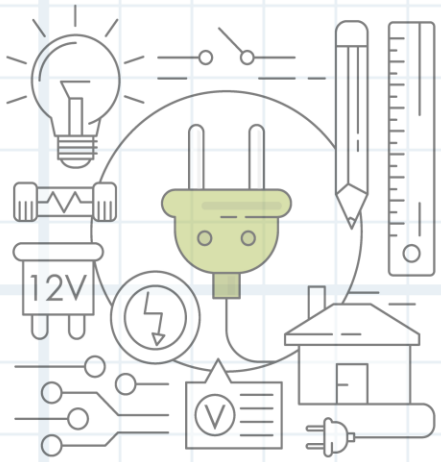


Islamic University Of Technology

Group-03

Course No: EEE-4418

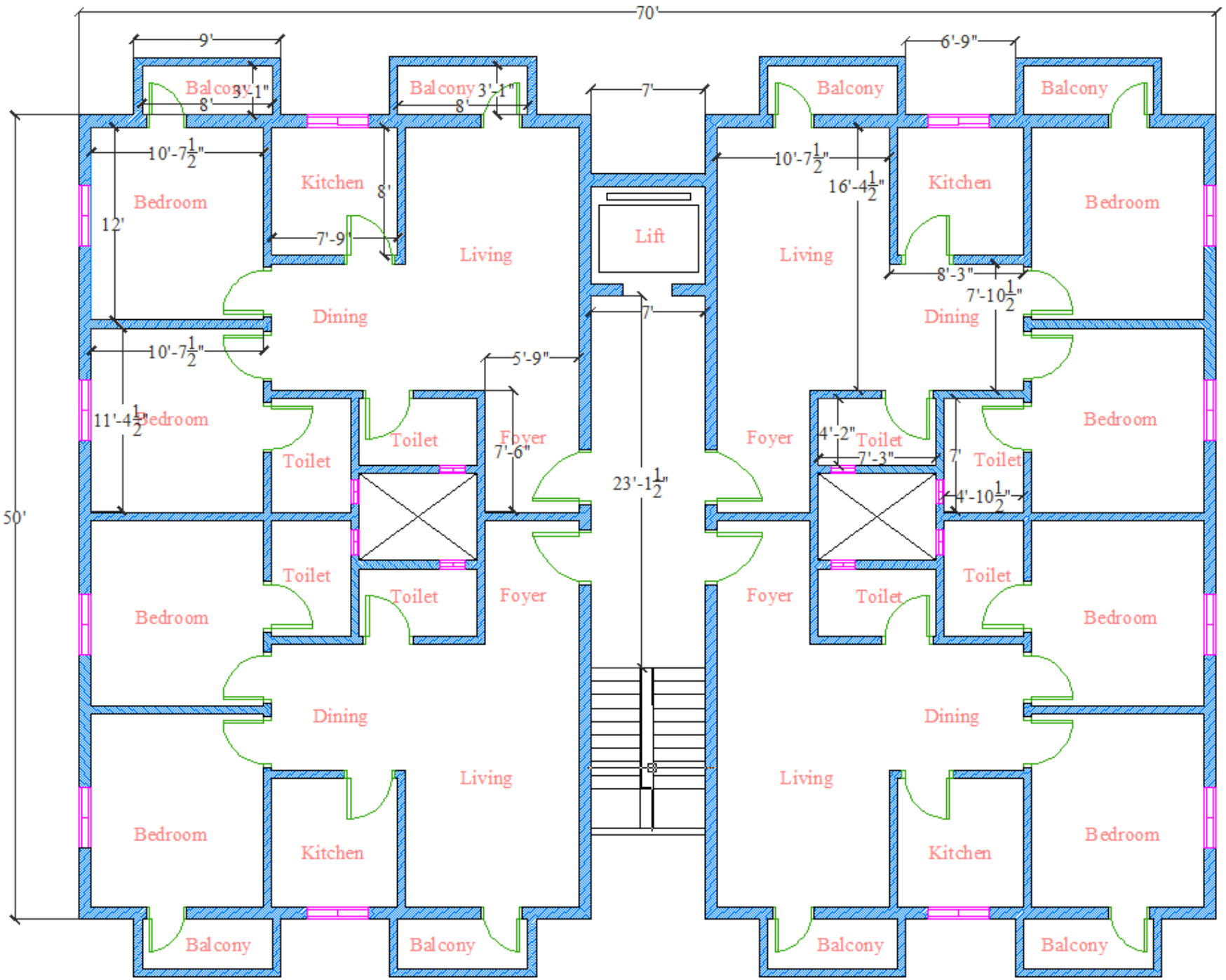
Topic: Structural Layout and Electrical Load Planning for Multistoried Building Using AutoCAD

Contributors:

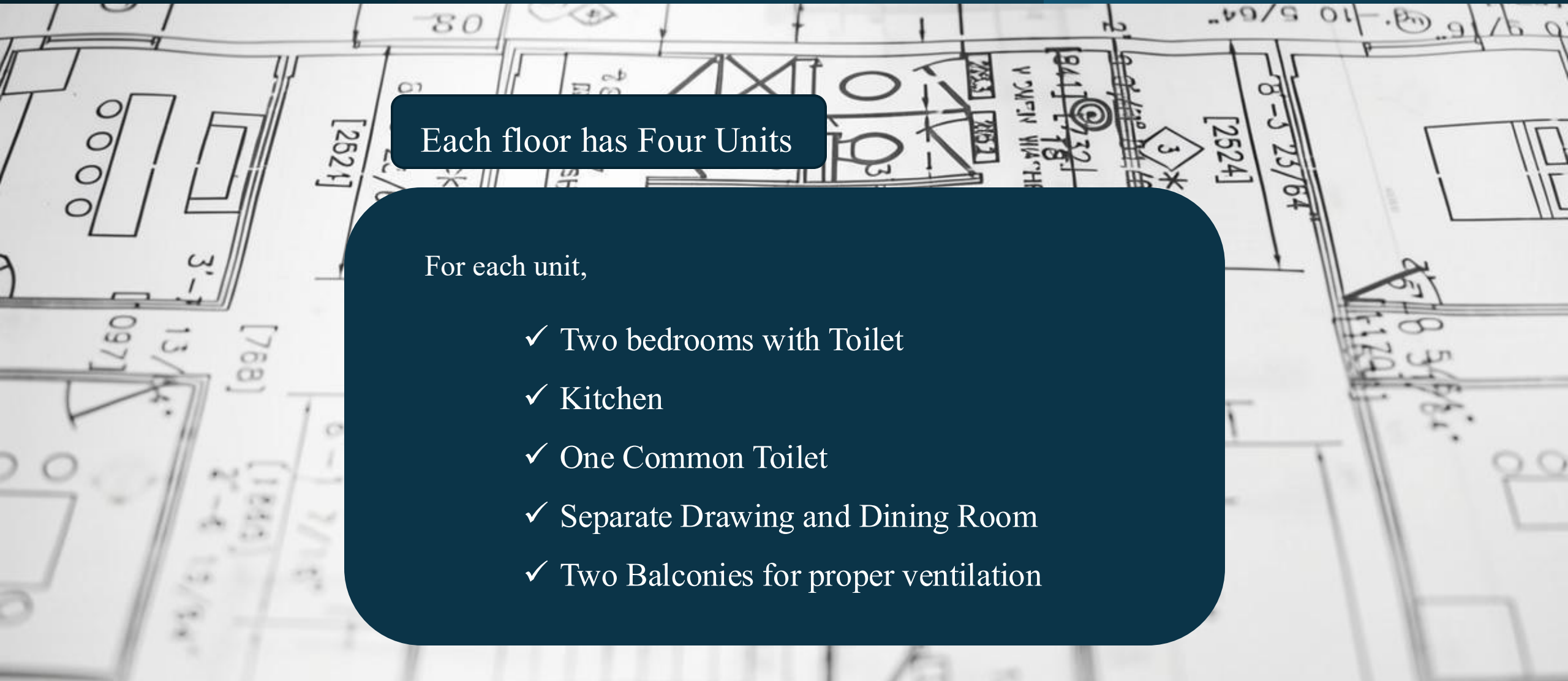
Name	ID
Md. Azmain Hasan	230021307
Fardin Amin Rafi	230021309
Sanajid Arham Bakar	230021311
Irfan Hossain Rafi	230021331
Mawanda Akram	230021349



Floor Plan:


70'
50'
9'
3'-1"
8'
Balcony
Balcony
3'-1"
8'
7'
6'-9"
Balcony
Balcony
10'-7½"
Bedroom
12'
Kitchen
8'
7'-9"
Living
Dining
Lift
10'-7½"
16'-4½"
Kitchen
Bedroom
Living
8'-3"
7'-10½"
Dining
7'
10'-7½"
5'-9"
11'-4½"
Bedroom
Toilet
Toilet
Foyer
7'-6"
23'-1½"
Foyer
4'-2"
Toilet
7'-3"
7'
Toilet
4'-10½"
Bedroom
Toilet
Toilet
Foyer
Foyer
Toilet
Toilet
Bedroom
Bedroom
Dining
Dining
Living
Living
Bedroom
Kitchen
Kitchen
Bedroom
Balcony
Balcony
Balcony
Balcony

Floorplan Description:

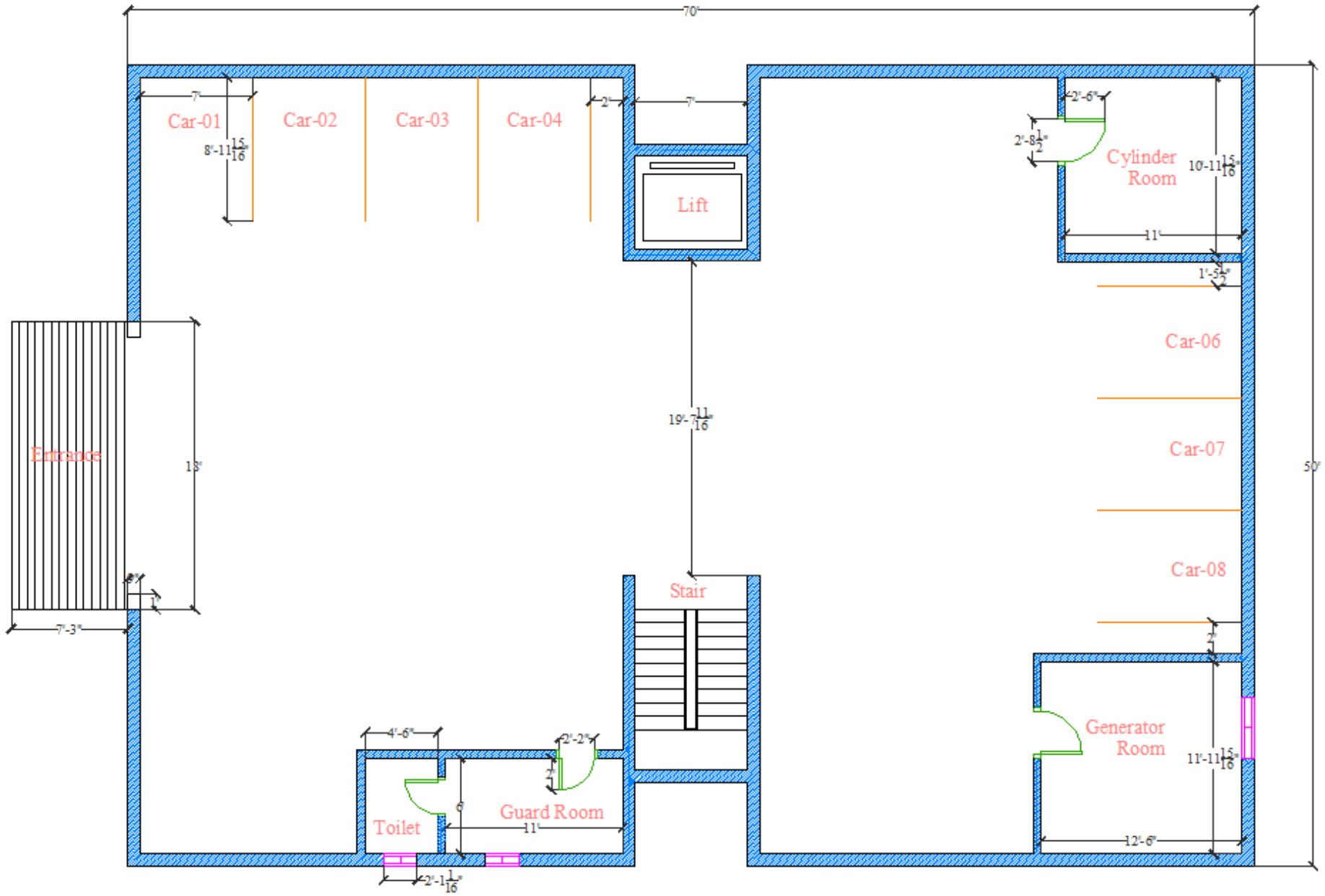


Each floor has Four Units

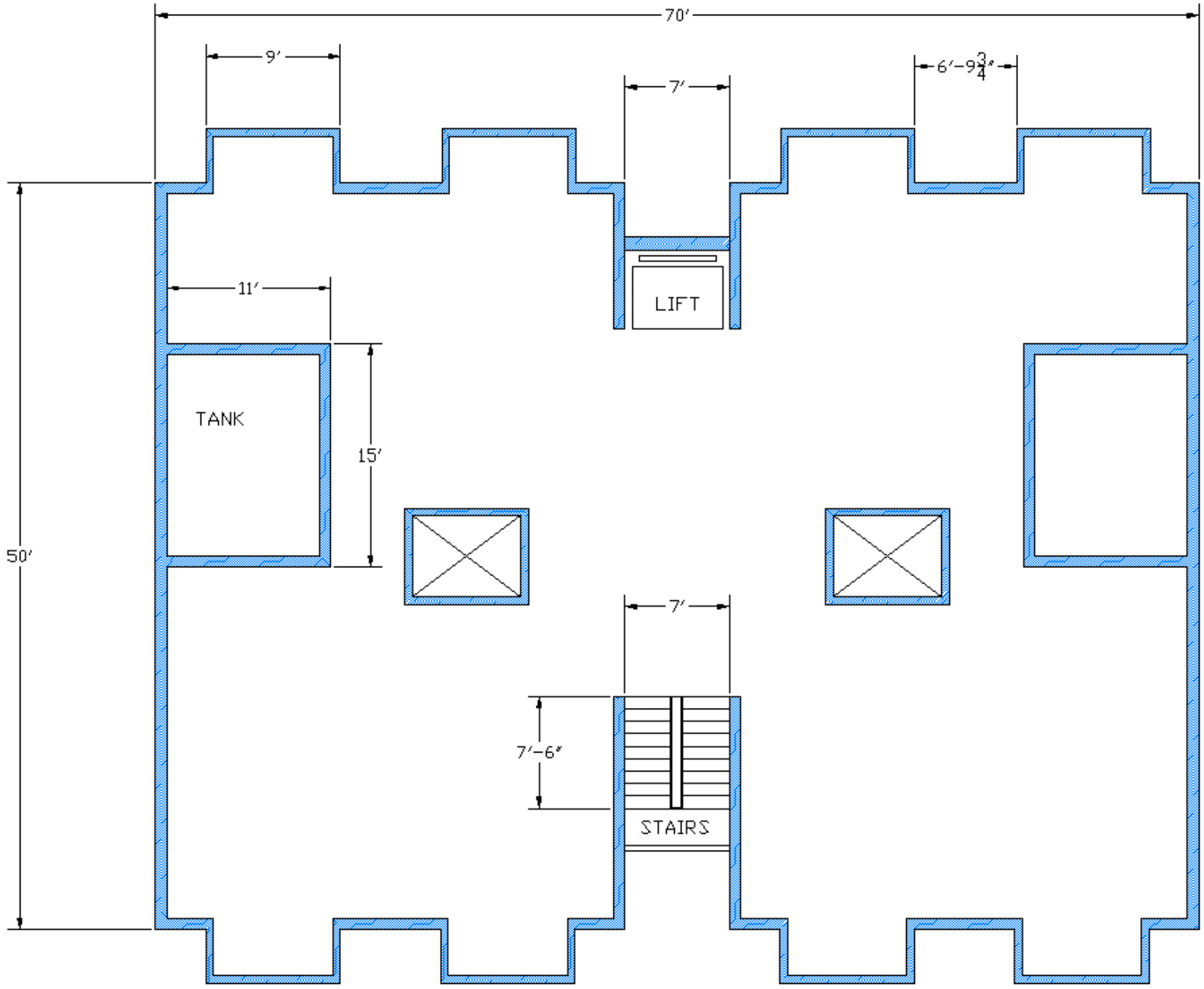
For each unit,

- ✓ Two bedrooms with Toilet
- ✓ Kitchen
- ✓ One Common Toilet
- ✓ Separate Drawing and Dining Room
- ✓ Two Balconies for proper ventilation

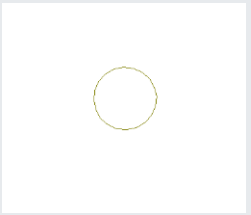
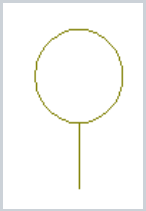
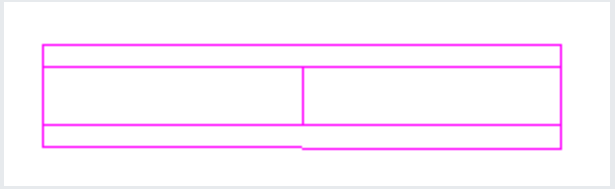
Ground Floor Plan:

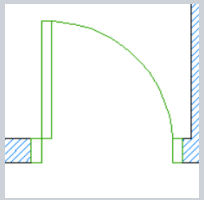
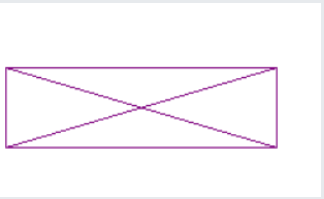
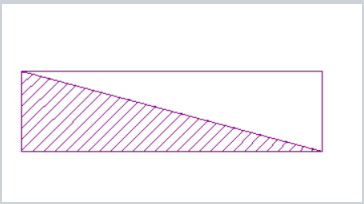


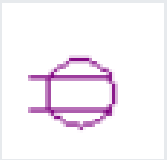
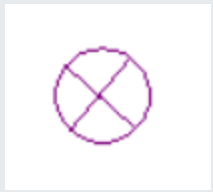
Rooftop Plan:



- **Symbols and other necessary descriptions:**

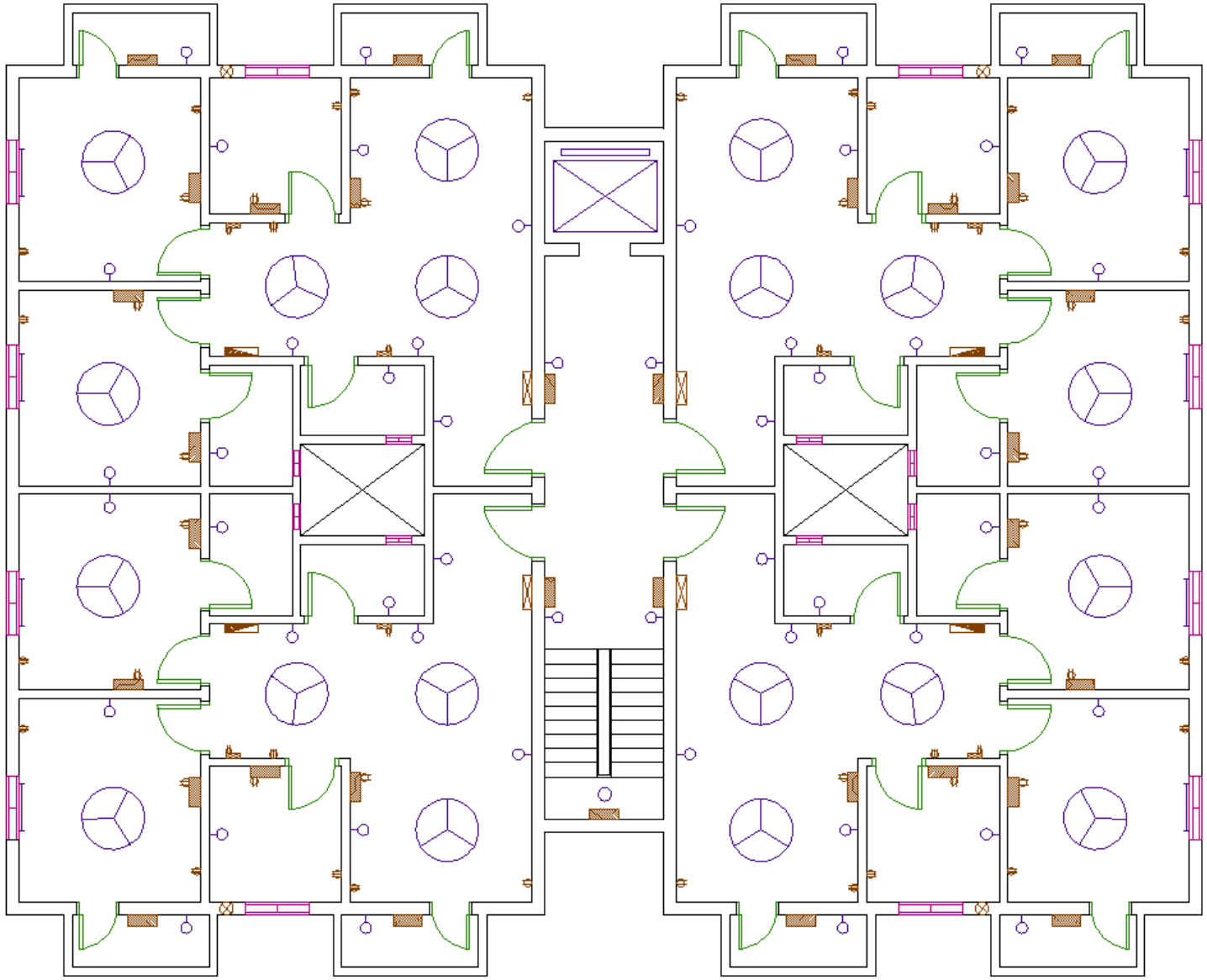
Symbol of Fittings and Fixtures	Name of the Components	Symbol Legend
	Tube Light	TB
	Light Bulb	LB
	Light Bulb	LB
	Window	W

	Fan	F
	Door	D
	Branch Distribution Board	BDB
	Emergency Sub-Distribution Board	ESDB

	Emergency Switch Board	ESB
	Three-pin socket	P
	Two-pin socket	S
	Switch Board	SB
	Exhaust Fan	EX

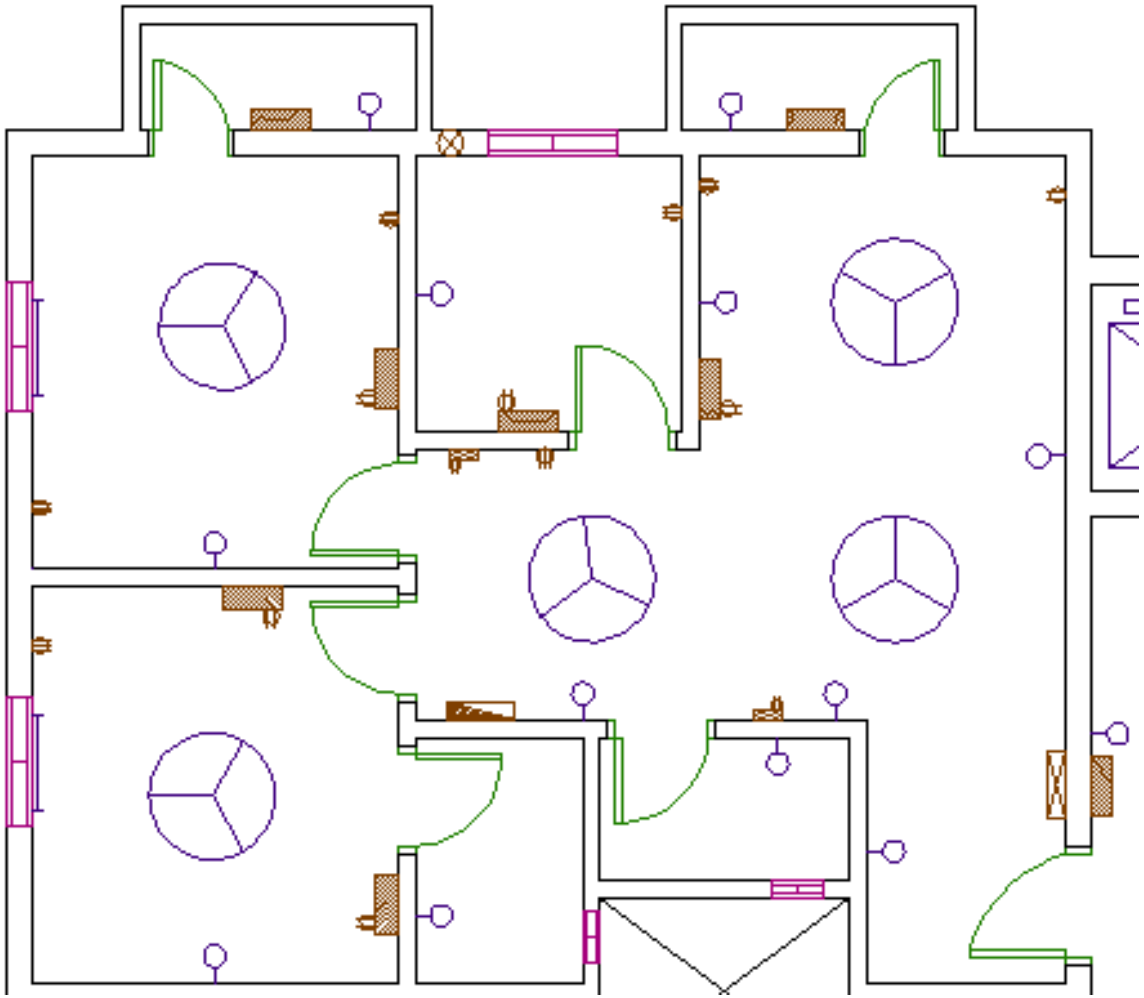
Apparatus	Total Number (Floor + Garage + Roof)	Rating
Fan	22	100 W
Light Bulb	82	20 W
Tube Light	8	20 W
2-pin socket	40	100 W
3-pin socket	25	3000 W
Lift	1	18.5 kW
Water Pump	1	5 kW
Generator	1	146 kW

- **Floorplan layout with fittings and fixtures:**

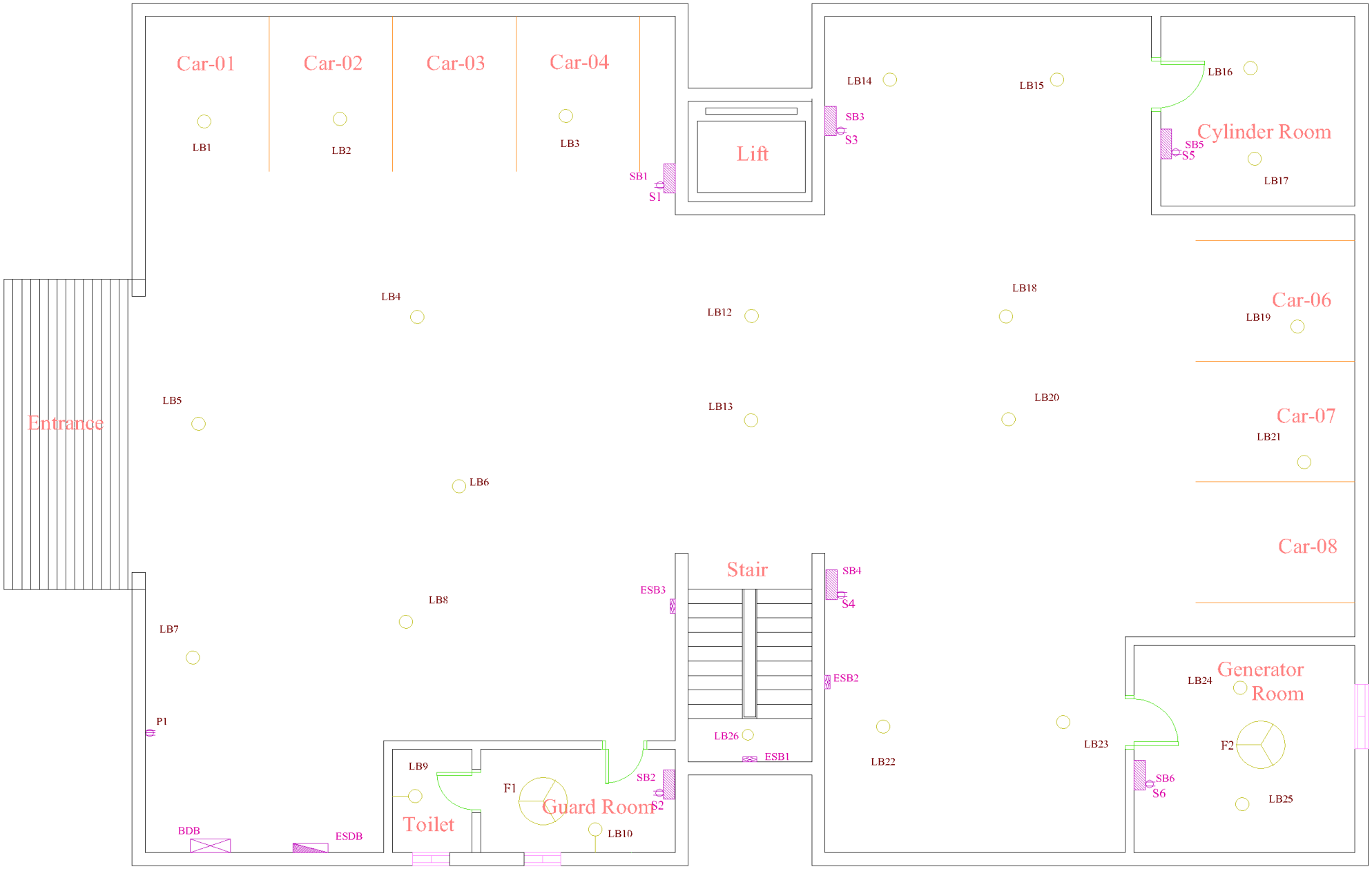


A typical unit of each floor

For better view:



Ground plan layout with fittings and fixtures:



➢Ground plan layout with fittings and Fixtures:

"No place will remain dark"

❖ **Generator Room:**

- Two lights, One fan and one two pin socket

❖ **Cylinder Room:**

- Two lights and One two pin socket

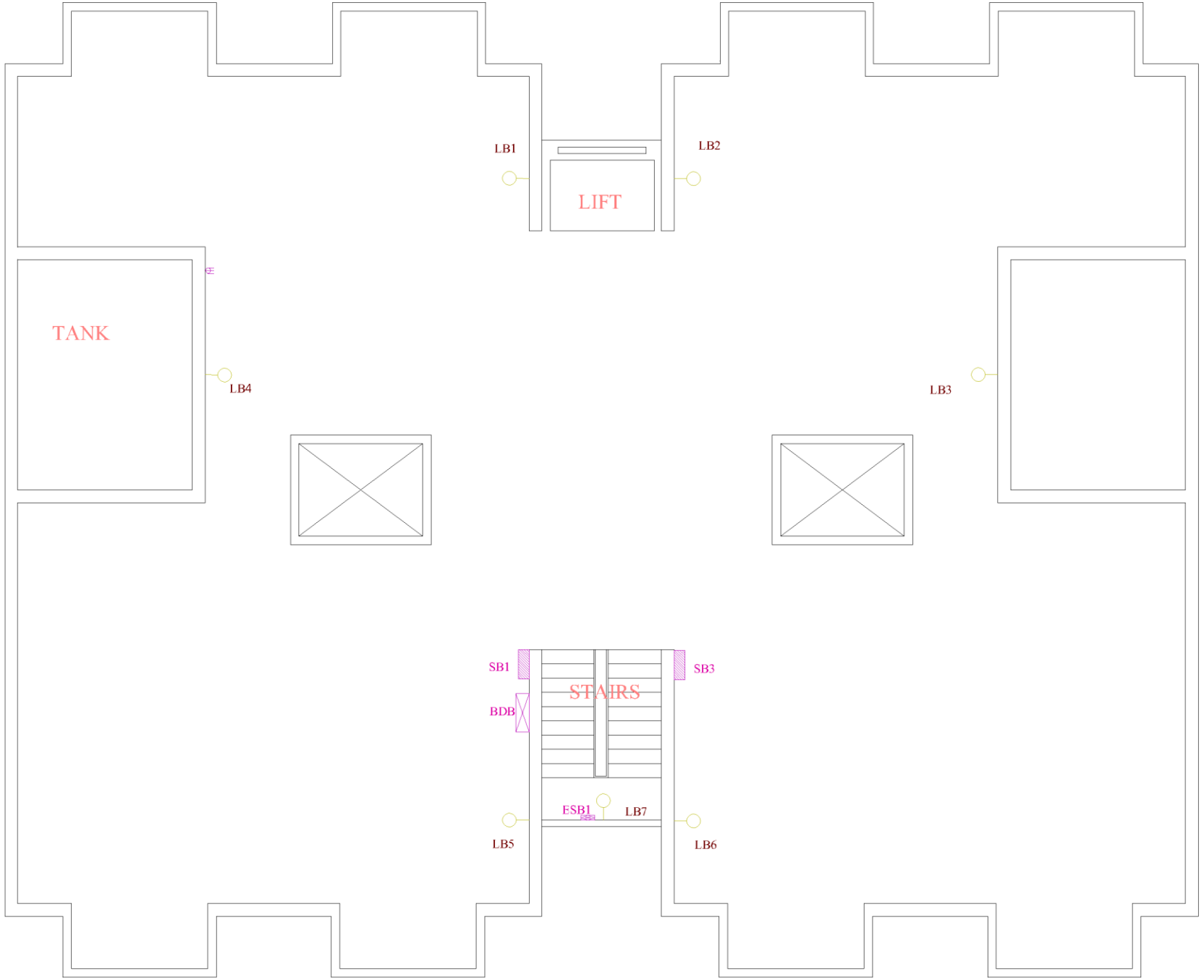
❖ **Guard Room:**

- Two lights, one fan and one two pin socket

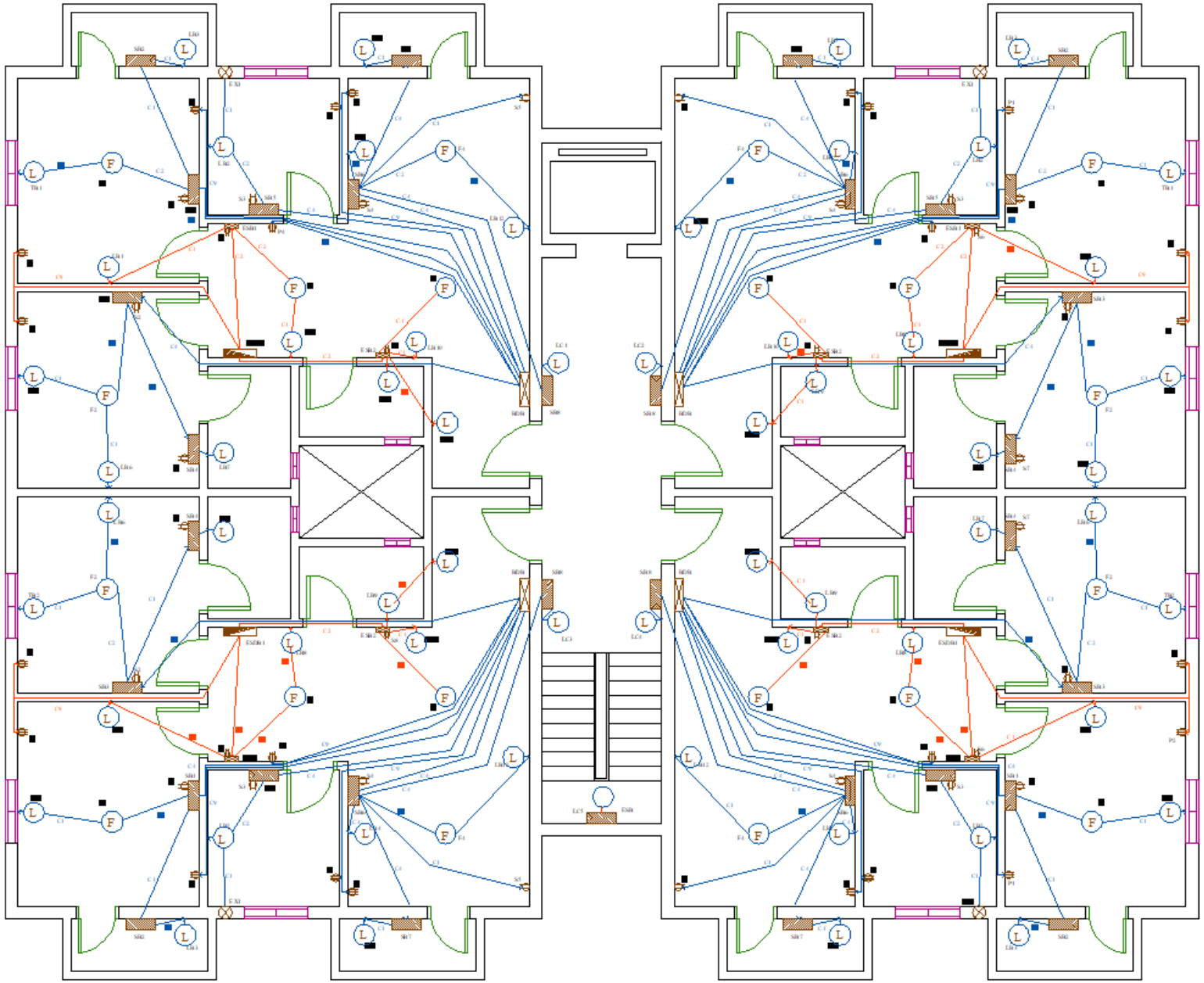
❖ **Parking area and the free spaces:**

- 18 lights, two pin socket and one three pin socket

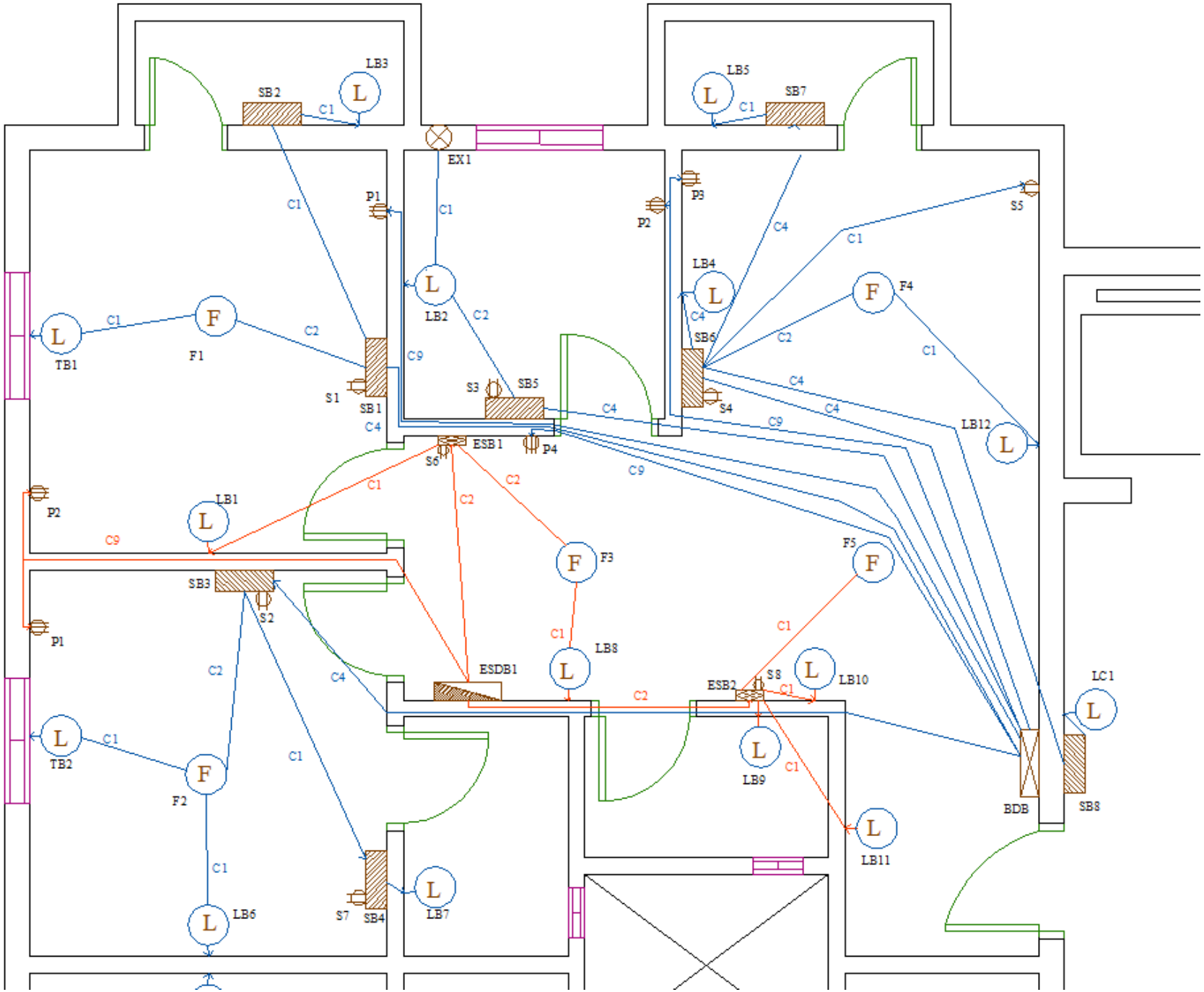
Rooftop Plan Layout with fittings and Fixtures:



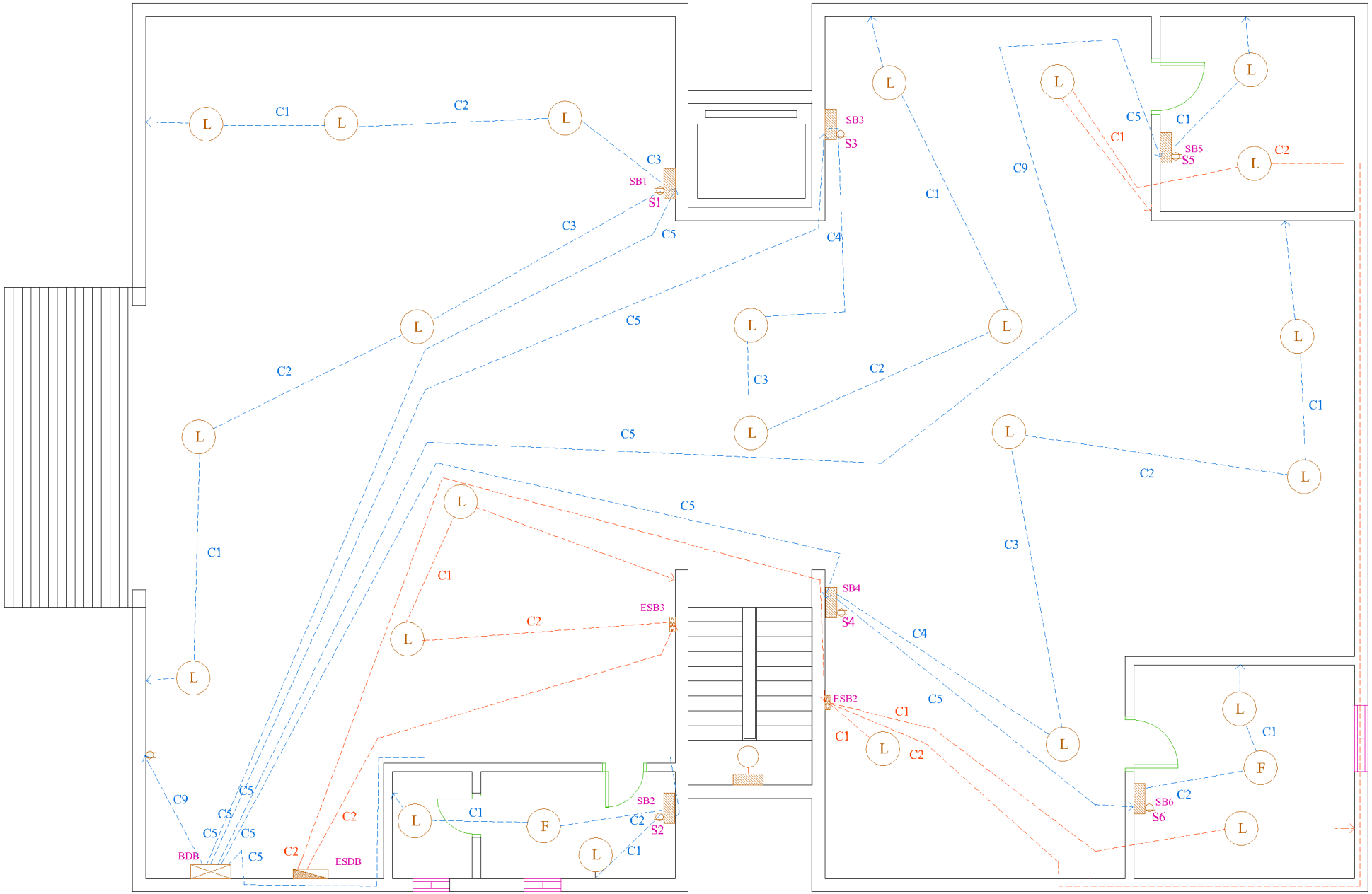
❑ Conduit Layout of Floorplan:



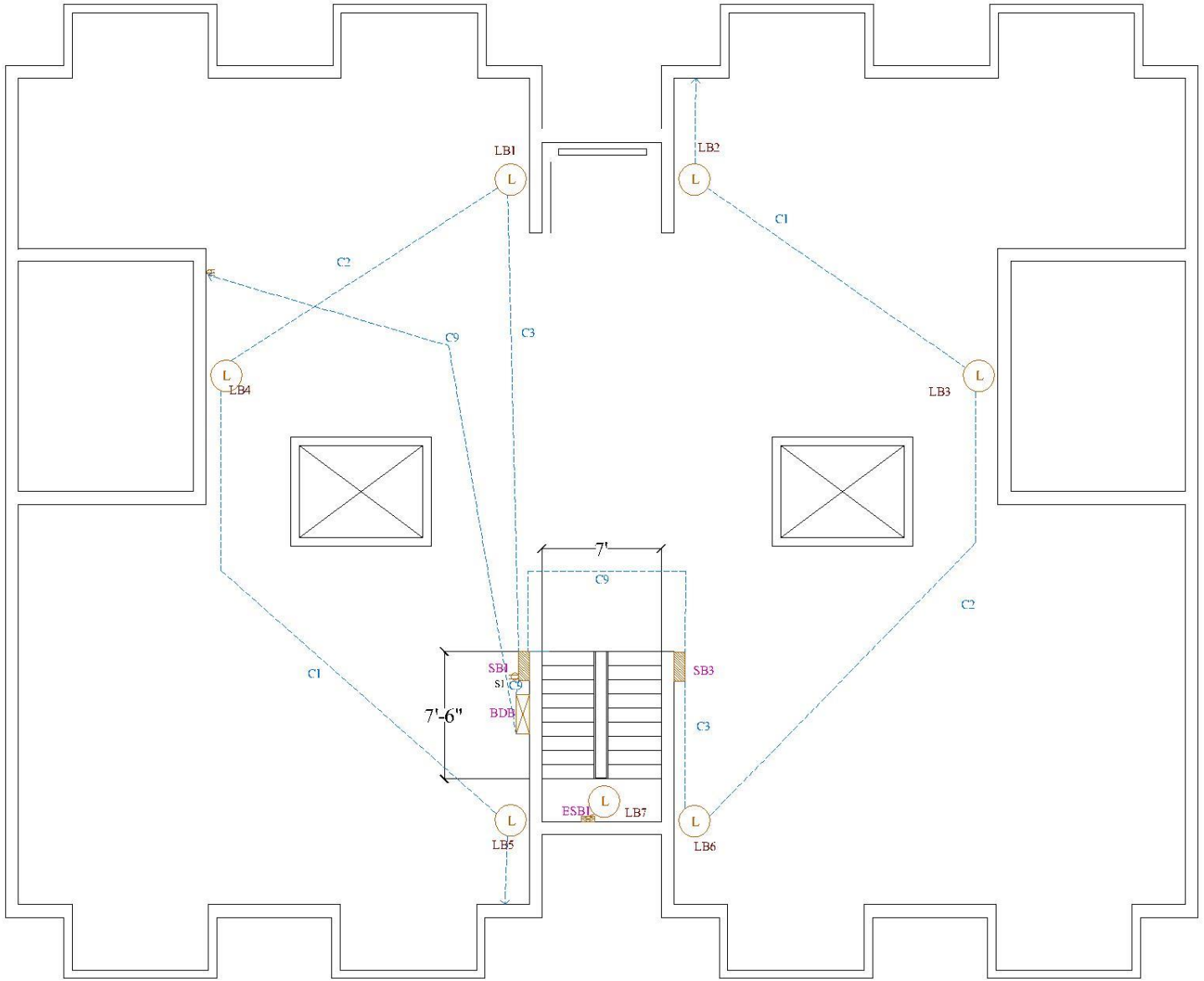
- **Better view:**


LB3
SB2
C1
LB5
SB7
EX1
P1
P2
P3
S5
LB2
LB4
F4
TB1
F1
S1
SB1
S3
SB5
SB6
S4
LB12
ESB1
P4
S6
P2
LB1
F3
F5
SB3
S2
P1
ESDB1
LB8
ESB2
S8
LB10
LC1
TB2
F2
LB9
BDB
SB8
LB11
LB6
S7
SB4
LB7

❑ Conduit Layout of Ground:



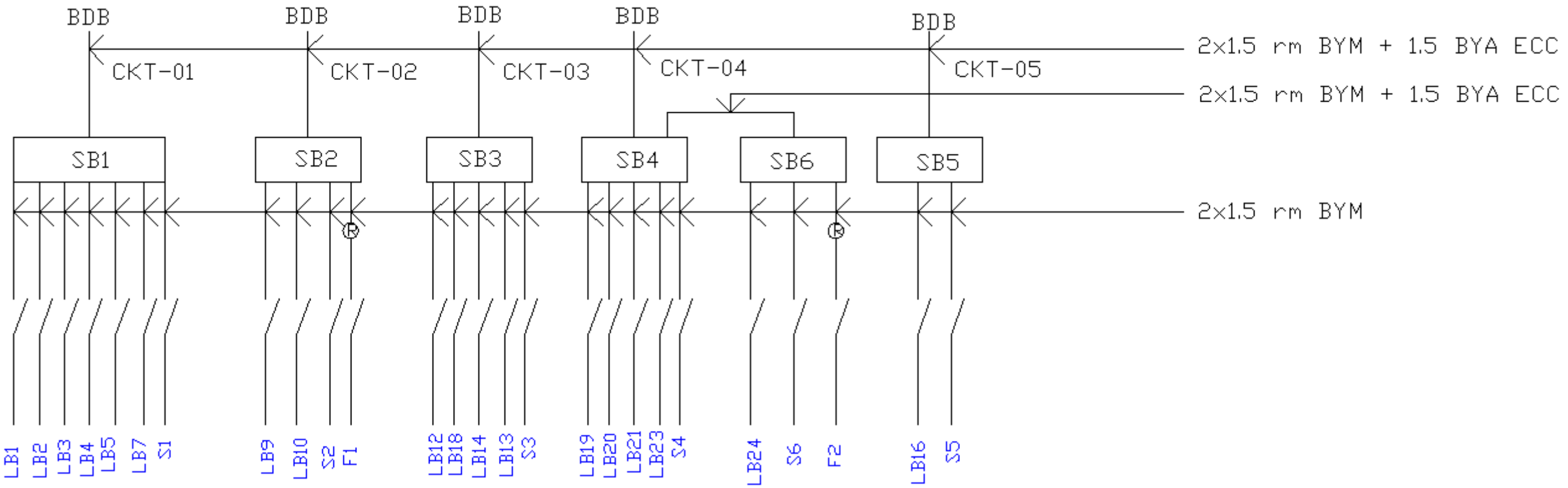
❑ Conduit Layout of Rooftop:



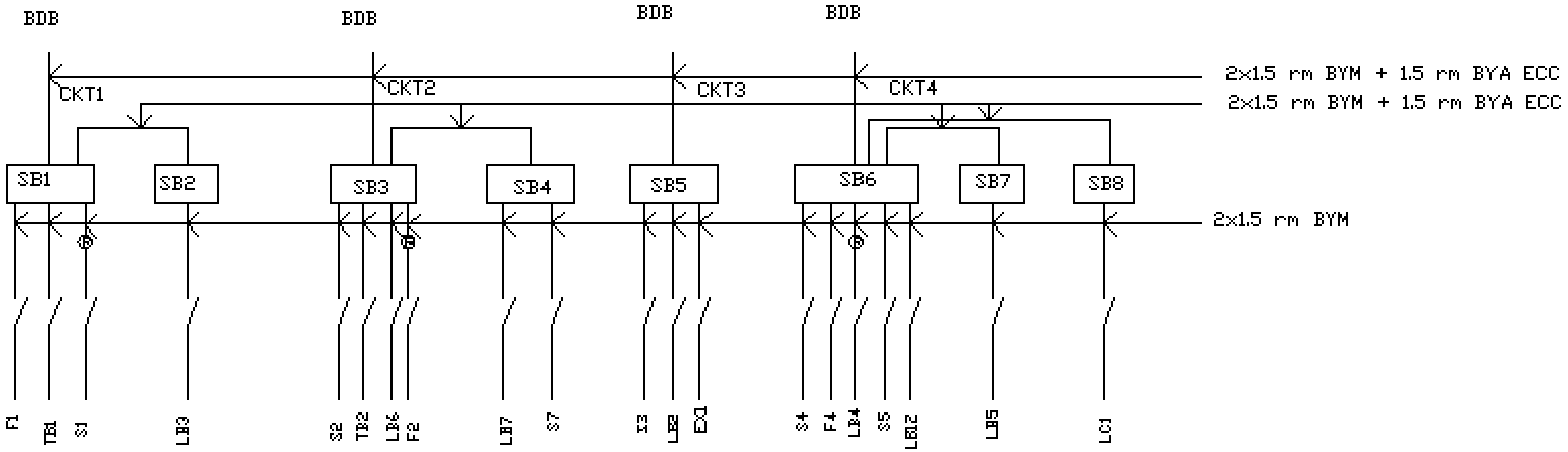
wire legend for your conduits:

C1	2 x 1.5 mm^2
C2	4 x 1.5 mm^2
C3	6 x 1.5 mm^2
C4	8 x 1.5 mm^2
C5	10 x 1.5 mm^2
C6	12 x 1.5 mm^2
C7	14 x 1.5 mm^2
C8	2 x 2.5 mm^2
C9	2 x 4 mm^2
C8,9	2 x 2.5 mm^2 and 2 x 4 mm^2

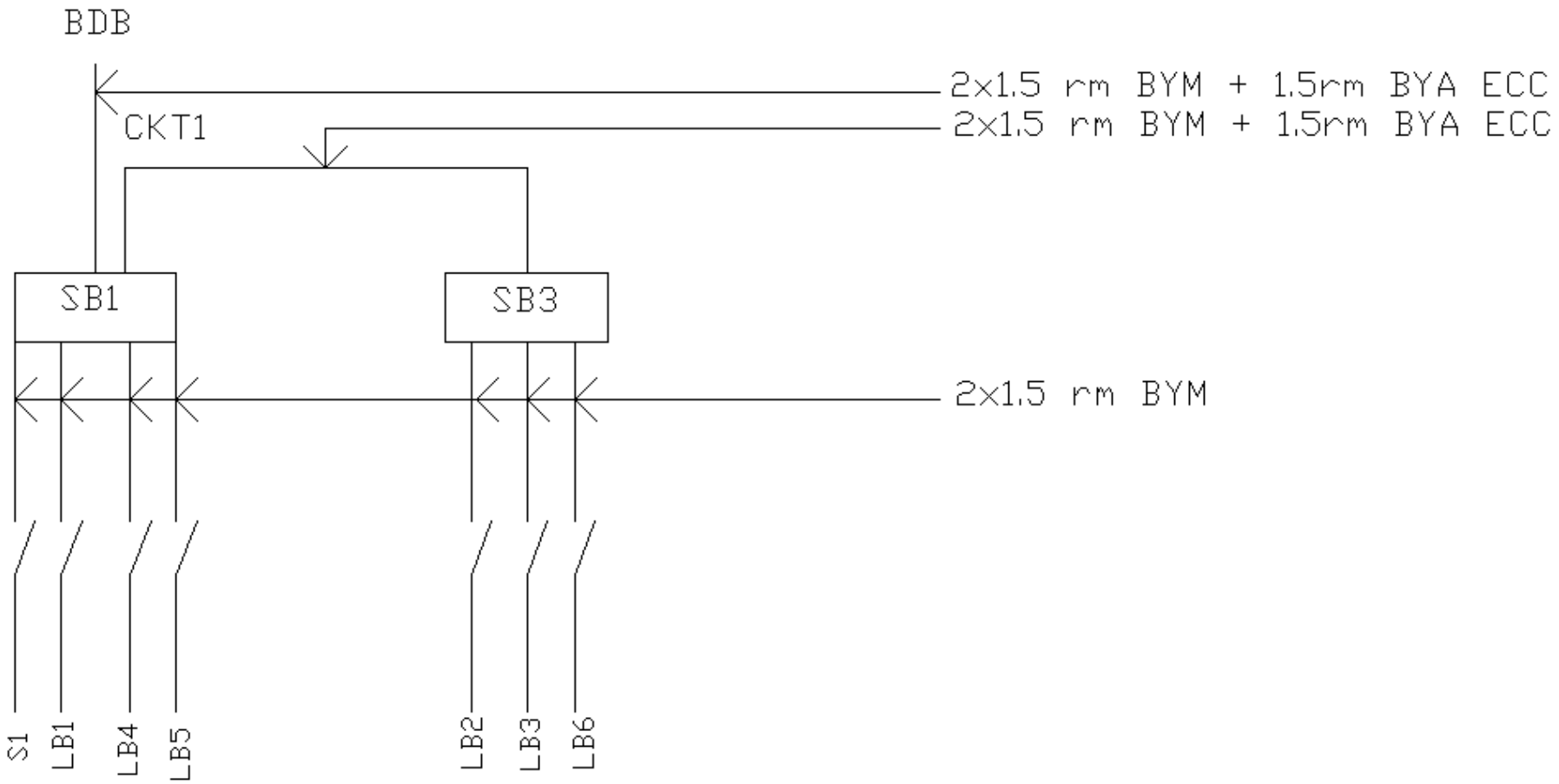
Switchboard Connection Diagram of Ground



Switchboard Connection Diagram for Each Floor



Switchboard Connection Diagram of Rooftop

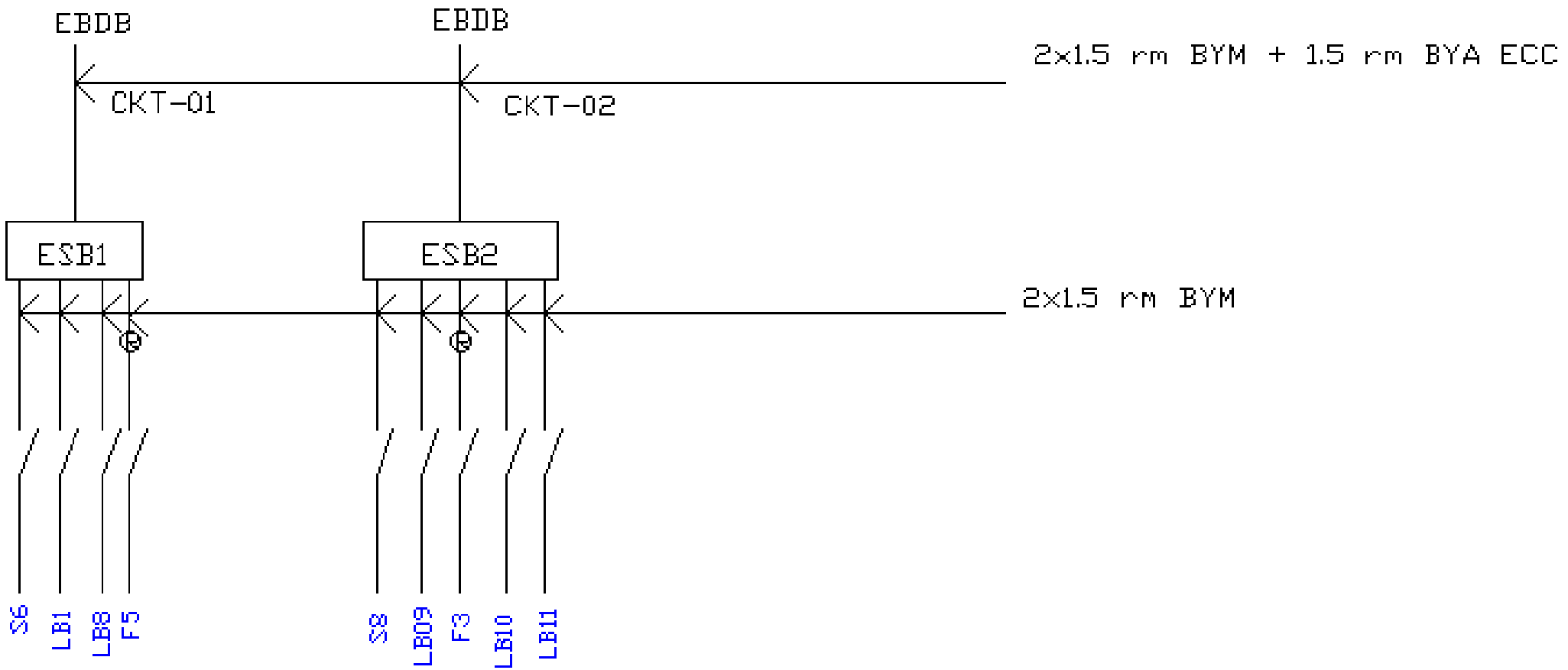


SELECTION OF CABLES & CIRCUIT BREAKERS SIZE

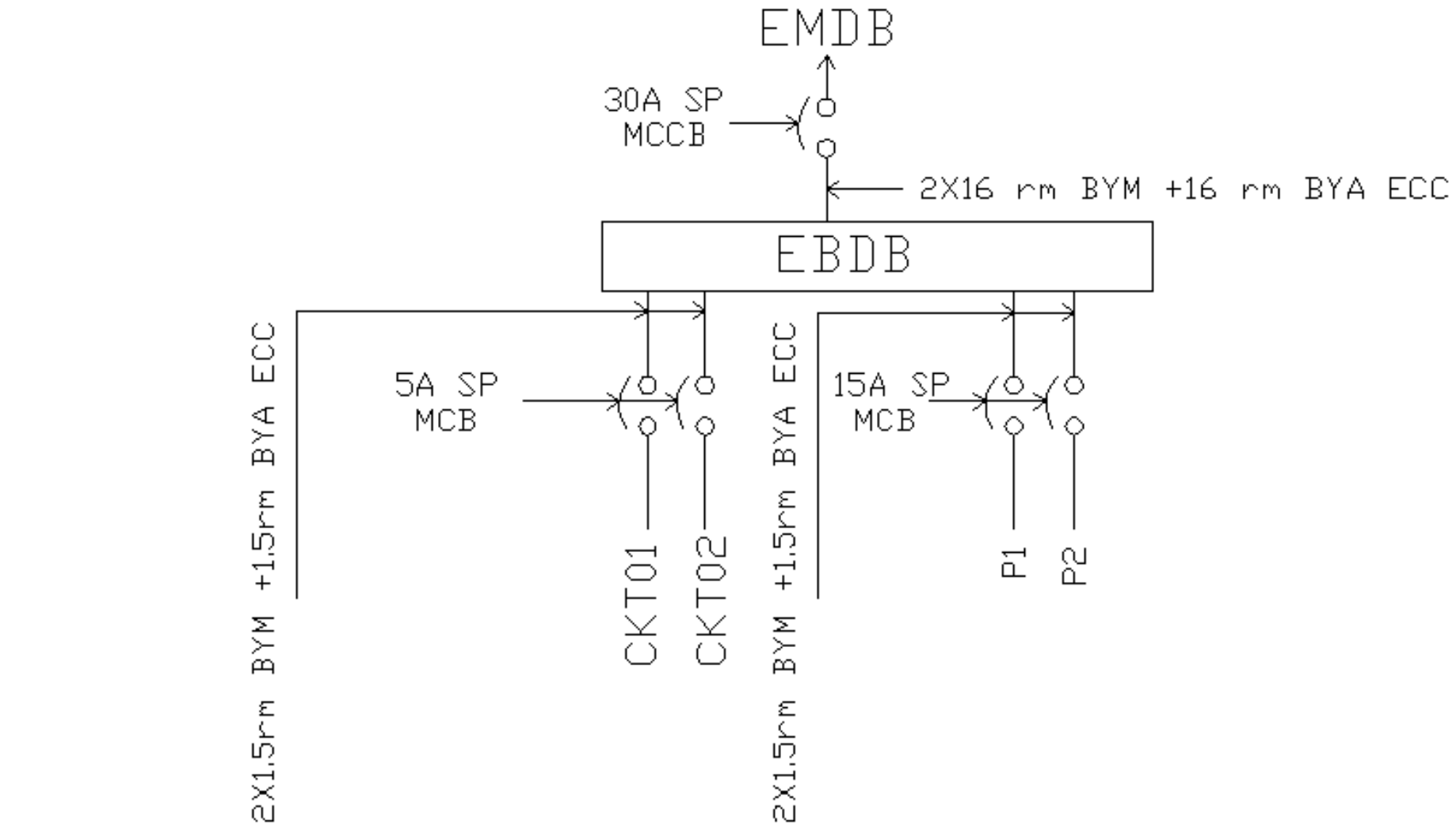
Table for Cables , Conduits , ECC , EL , Voltage drop and Current ratings of different specifications as per Manual of Eastern Cables , BICC cables and Tables , Electrical Conductors (International Standard Sizes) etc. :

A	B	C	D	E	F		G	H	I		J	
					a'	b'			a"	b"	a"'	B"'
3/0.029	1.5	5	16	10	6	10		27	27	22	16	20
7/0.029	2.5	10	16	10	4	7		16	36	30	22	28
7/0.036	4	15	14	10	3	5	1	10	47	39	30	37
7/0.044	6	20	14	10	2	4	1	6.8	59	50	38	47
7/0.052	10	30	10	10	1	2	1.5	4	78	68	52	63
7/0.064	16	40	10	10		1	1.5	2.6	100	94	70	85
19/0.052	25	50	6	6		1	2	1.6	130	125	91	110
19/0.064	35	60	6	6			2	1.2	155	160	112	136
19/0.072	50	70	6	6			2	0.93	185	195	136	164
19/0.083	70	100	1/0	1/0			2	0.65	225	245	173	207
37/0.072	95	120	1/0	1/0			2.5	0.48	270	300	216	253
37/0.083	120	150	1/0	1/0			2.5	0.4	310	350	244	291
37/0.093	150	200	1/0	1/0			3	0.34	350	405		333
37/0.130	185	250	3/0	3/0			3.5	0.29	390	460		381
61/0.093	240	300	3/0	3/0			4	0.24	450	555		452
61/0.103	300	425	3/0	3/0			4	0.22	515	640		526
91/0.093	400	585	3/0	3/0			6	0.2	586	770		639
91/0.103	500	685	3/0	3/0			6	0.18	680	900		752
127/0.103	630	800	3/0	3/0			6	0.17	800	1030		855

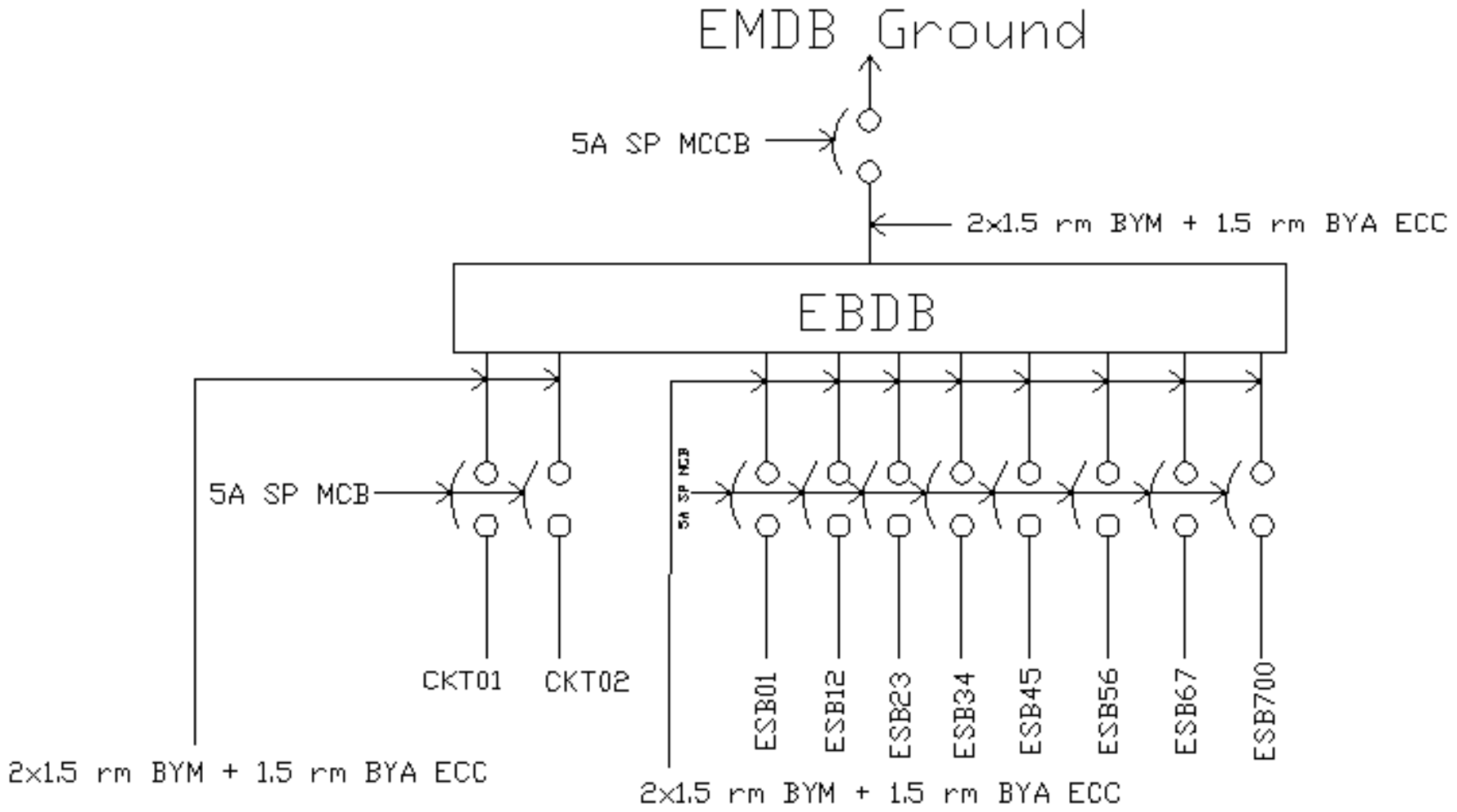
EBDB Connection Diagram of Floorplan



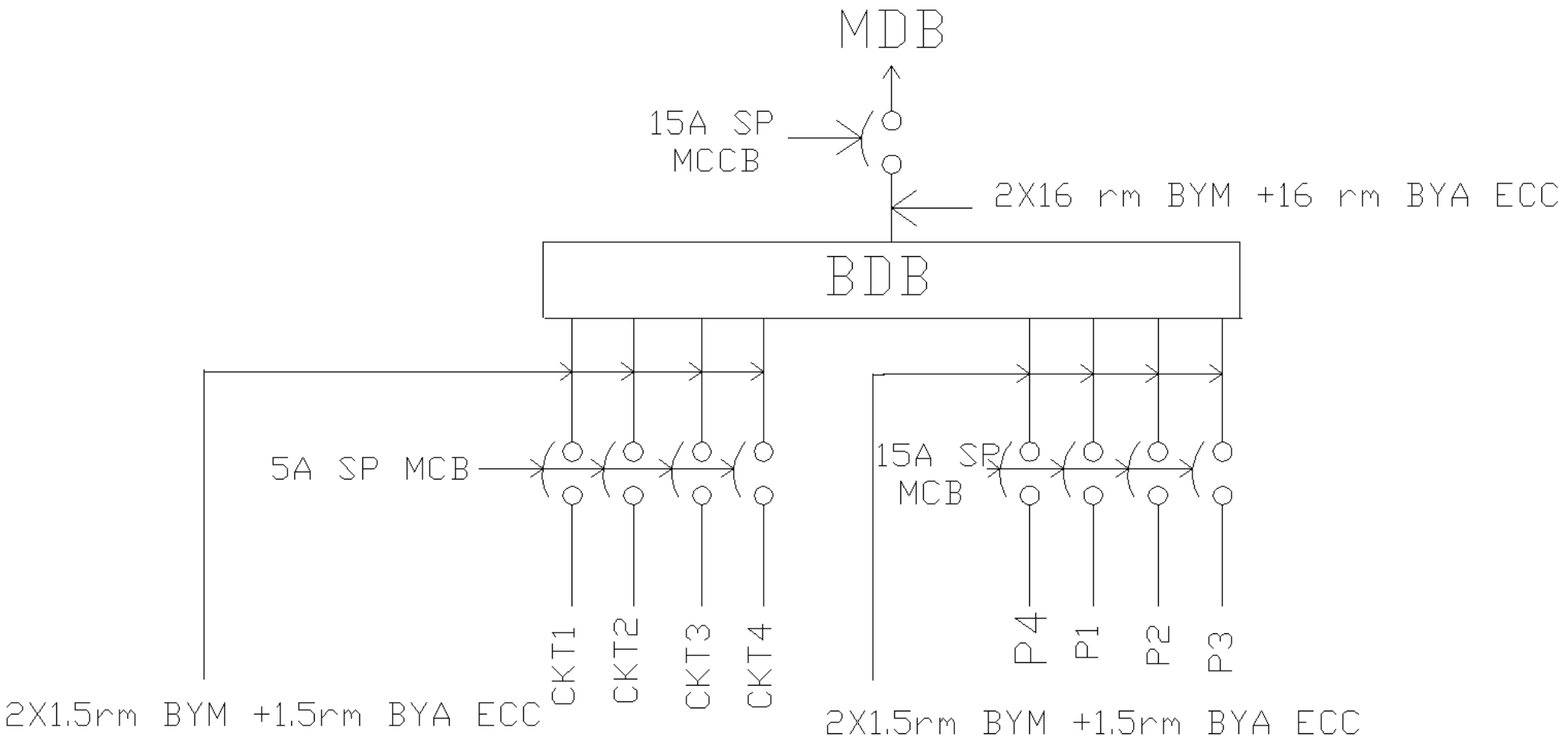
EBDB Connection Diagram of floor with EMDB



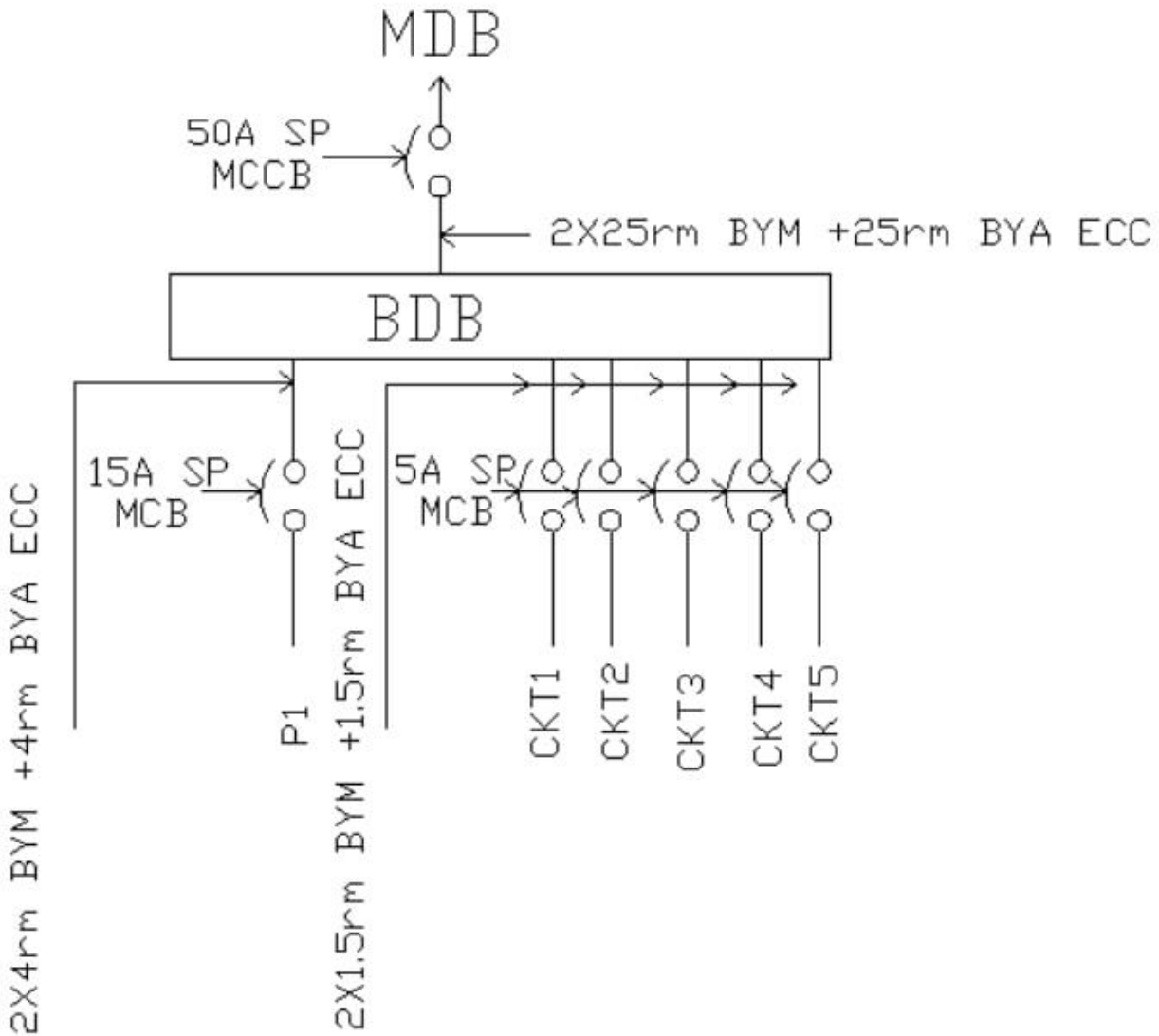
EBDB Connection Diagram of Ground and Stairs with EMDB



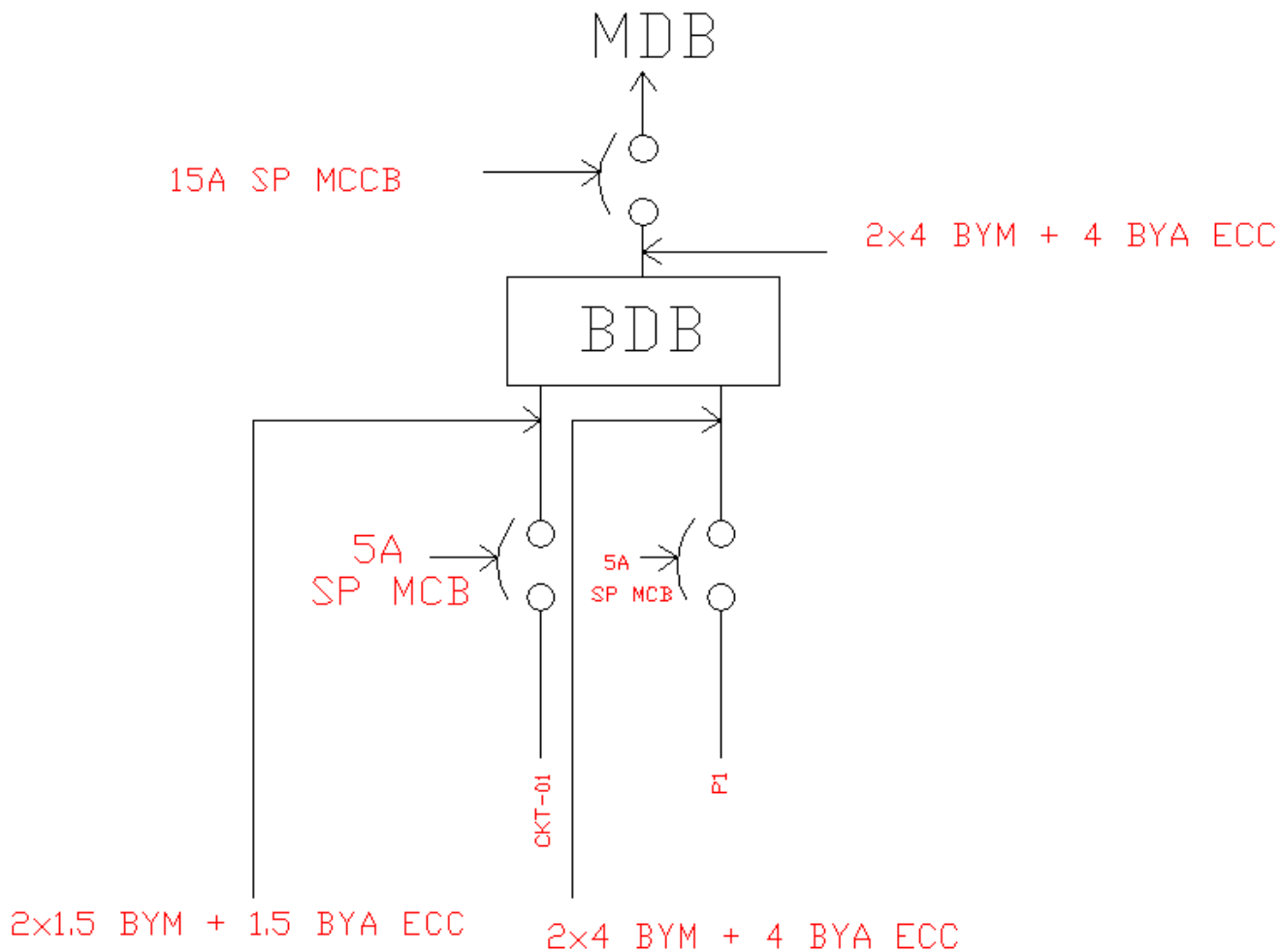
MDB Connection Diagram of Each Floor



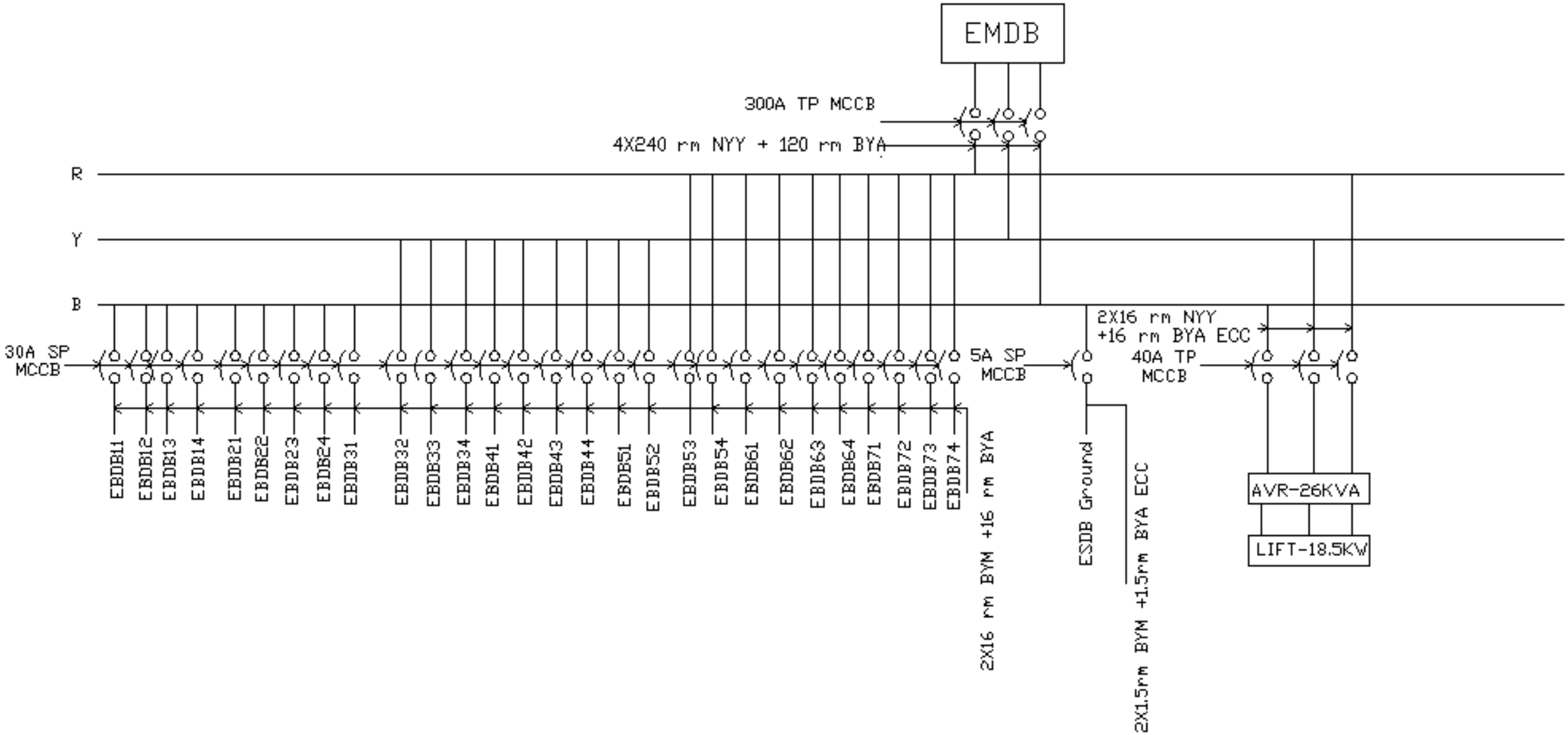
- **MDB Connection Diagram of Ground:**



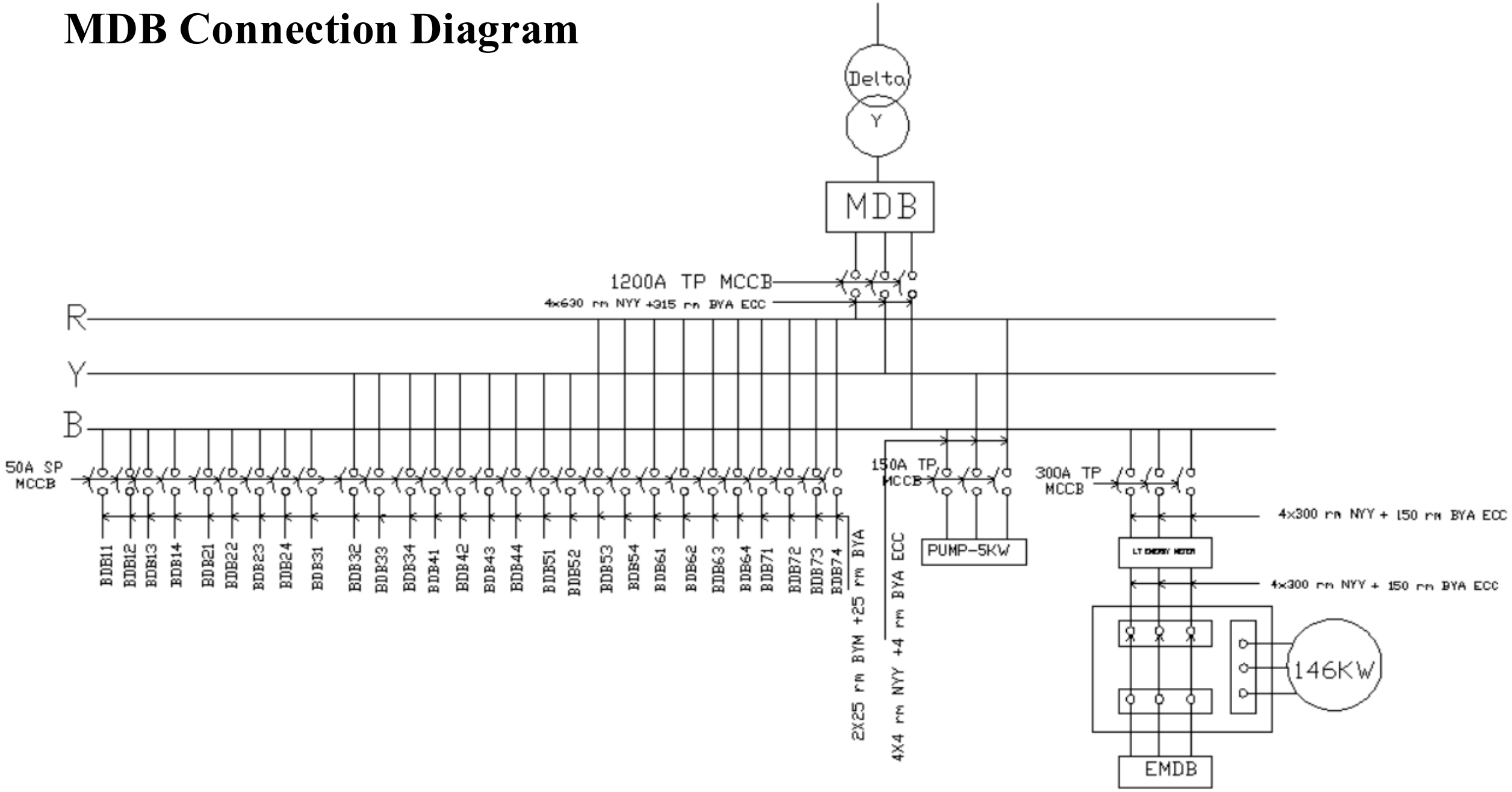
- **MDB Connection Diagram of Rooftop:**



EMDB Connection Diagram:

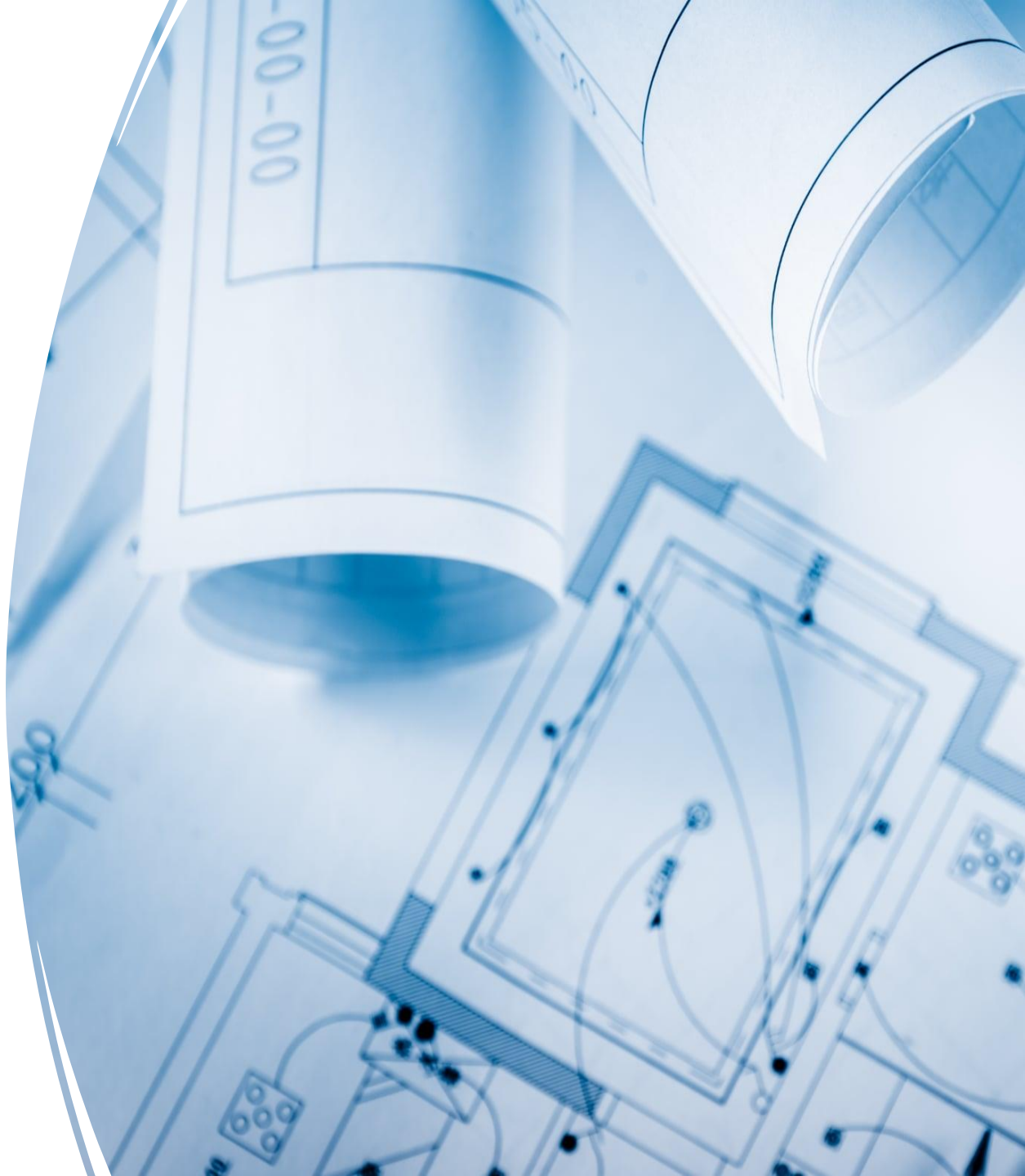


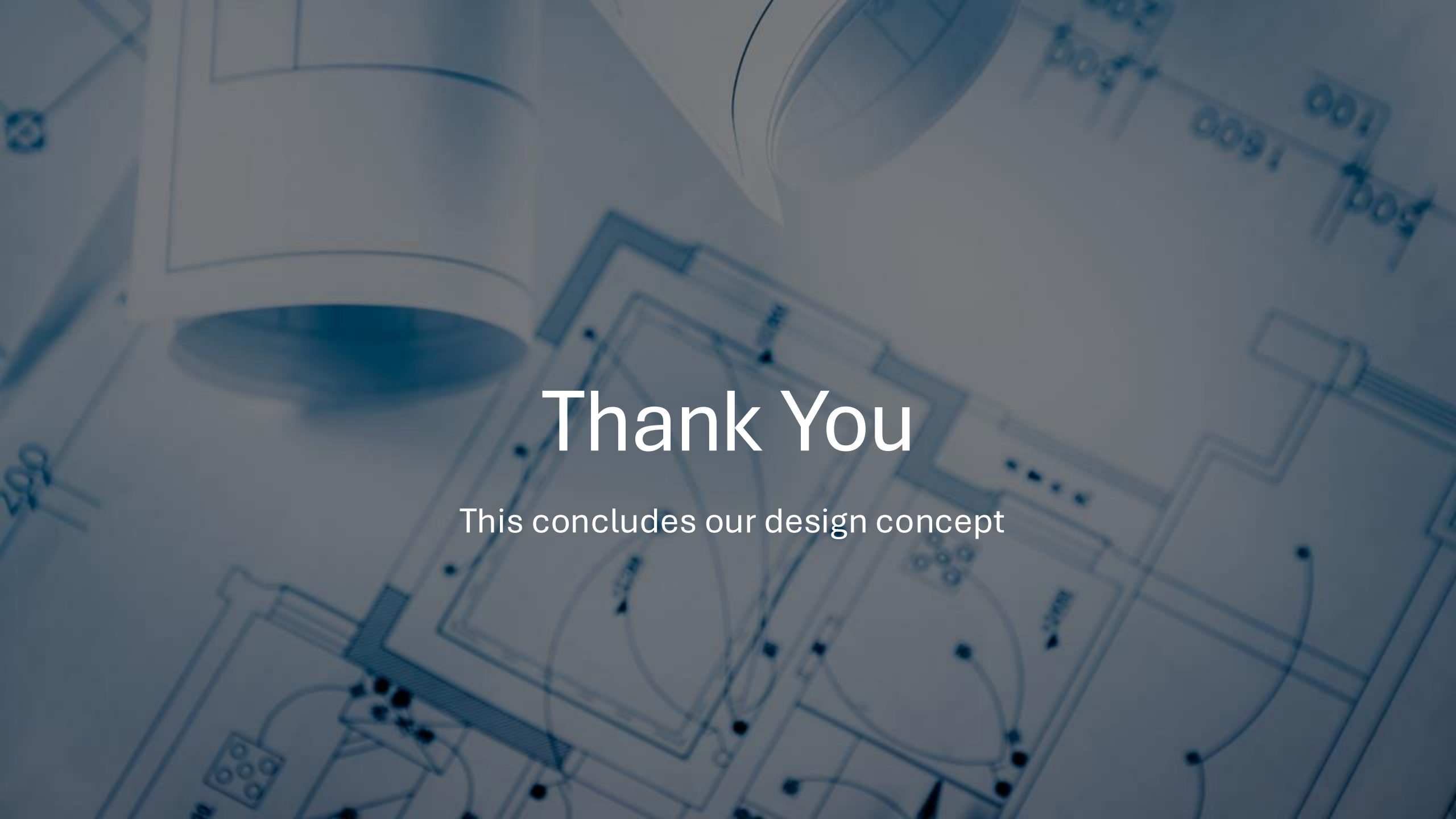
MDB Connection Diagram



What Makes Our Design Plan Stand Out

- **Symmetrical Layout:** 4-quadrant mirrored units around a central lobby
- **Open Utility Space:** Flexible layout suited for parking, security, and main service connections.
- **Two Open Spaces:** Attached to washrooms which allows for ventilation
- **Two Water Tanks:** If one breaks down, the other can still supply water as backup, ensuring water security.





Thank You

This concludes our design concept